Cross-Scale Remote Sensing Estimating of Winter Wheat Above-Ground Biomass (AGB) From UAV to Satellite: Enhanced by Phenological Matching and Data Transfer

Yongji Zhu, Yinglun Li, Xinyu Guo, Jinling Zhao, Xinwei Li, Jikai Liu, Yanli Chen, and Linsheng Huang

***Abstract*—Accurately assessing the above-ground biomass (AGB) in the winter wheat is essential for evaluating the crop development and ensuring the food security. Data-driven model transfer from the unmanned aerial vehicle (UAV) to satellite typically demands the extensive UAV data and often exhibits limited generalizability across years and regions due to spatiotemporal heterogeneity in crop growth dynamics. Although phenology-driven models offer the potential for continuous estimation of crop growth parameters, significant phenological variations often exist across large regions, resulting in the poor model transferability. To address these challenges, this study proposes a cross-scale AGB estimation framework that integrates the phenological matching with data transfer (PDBM) approach. Leveraging time-series vegetation index (VI) data from both UAV and satellite, PDBM performs the spatiotemporal phenological matching and couples it with data transfer to enable the upscaling of AGB models from UAV to satellite scales. At the UAV scale, using the wide dynamic range VI (WDRVI) as an example, PDBM significantly improved the estimation accuracy across years compared with data-driven model (R^2: 0.79 versus 0.86 for 20212022 and 0.85 versus 0 for 2023) and also outperformed phenology-driven model (R^2: 0.79 versus 0.79 for 2021 and 2022 and 0.85 versus 0.78 for 2023). At the satellite scale, using the soil-adjusted VI (SAVI) as an example, PDBM also outperformed a data-driven model (R^2: 0.79 versus 0 for 2010) and a phenology-driven model (R^2: 0.79 versus 0 for 2010). Results show that PDBM enables effective upscaling of AGB models from UAV-to-satellite scales and accurately captures both the spatial patterns and interannual variability of winter wheat AGB. In summary, the proposed PDBM framework holds great promise as a robust tool for precision agriculture.**

***Index Terms*—Above-ground biomass (AGB), cross-scale remote sensing, phenology, Satellite, unmanned aerial vehicle (UAV).**

I. Introduction

WINTER wheat, a key cereal crop worldwide, plays a crucial role in temperate climates [1]. Above-ground biomass (AGB) is not only a crucial indicator for assessing crop growth [2] but also an essential factor in yield estimation [3], [4]. The accurate estimation of winter wheat biomass is crucial for optimizing agricultural decision, implementing precision agriculture practices, and ensuring food security [5]. However, traditional estimating techniques, such as field sampling, although highly accurate, are resource-demanding, time-consuming, and expensive, rendering them impractical for large-scale and real-time estimating needs [6], [7].

The rapid development of remote sensing technologies has introduced novel solutions for estimating crop biomass. Compared with traditional methods, remote sensing greatly improves efficiency thanks to its extensive coverage, frequent updates, and noninvasive approach [8], [9]. Among various remote sensing platforms, both unmanned aerial vehicles (UAVs) and satellites offer unique advantages. UAVs are ideal for detailed estimation and model development at the field scale due to their high-spatiotemporal resolution and operational adaptability [10], [11], while satellites, with their expansive coverage, enable estimation of crops at regional and global levels [12], [13]. In recent years, multiplatform collaborative estimating approaches have become increasingly popular [14], [15], [16], [17]. However, factors, such as spatiotemporal resolution, limit the effectiveness of multiplatform crop estimation. Research by Chen et al. [18] and Hu et al. [19] has successfully addressed the discrepancy between the ground sampling and satellite resolution by using UAV-derived data to calibrate the satellite imagery. Nevertheless, the high cost of acquiring data and the complexity of processing workflows hinder the widespread application of UAVs in large-scale satellite upscaling [20].

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Therefore, overcoming data acquisition and processing bottlenecks and developing efficient scaling models from UAV to satellite platforms have emerged as critical scientific challenges in the estimation of agricultural remote sensing [21].

At present, estimating winter wheat parameters using remote sensing mainly relies on data-driven models [22], but it faces two significant challenges. First, machine learning and data mining algorithms usually assume that the feature space and distribution are consistent in both the training and testing datasets [23]. However, changes in growth stages, varieties, field management, and environmental conditions can lead to significant shifts in feature distribution, reducing the model's transferability. Some studies have attempted to enhance the estimation ability of models through data transfer. For example, Amitrano et al. [24] combined active learning strategies with regression models to minimize on-site sampling and estimate vegetation biomass and nitrogen content. Liu et al. [25] used a combination of active learning optimization and spectral angle mapper (SAM) methods to select PROSAIL-PRO simulation samples that closely matched real-world conditions. They also assigned geographical location information to the samples in order to estimate the AGB of rice in situations where the sample size was small. Wang et al. [26] improved the estimation of spatiotemporal shoot P uptake (SPU) at a regional scale using an active learning algorithm in cases of sparse data. However, these methods primarily depend on updating and optimizing the dataset to enhance the model's generalization. Second, traditional models often overlook the dynamic influence of phenological stages on biomass [27]. Models developed based on a specific growth period are frequently impacted by changes in phenological stages when applied [28], [29]. For example, the correlation between vegetation indices (VIs) and biomass varies considerably during the jointing and heading stages. Ignoring the effects of phenology can result in a decline in model accuracy [30]. Therefore, it is necessary to consider the influence of phenological effects on the model.

Phenological detection methods based on VI time-series offer novel solutions to the aforementioned challenges [31]. Existing approaches include threshold-based techniques [32], function-fitting methods [33], and shape model fitting (SMF) methods [34]. The crop phenology can be relatively easily detected using preset thresholds. However, when setting absolute or relative thresholds for different crop phenological stages, environmental differences between locations must be considered, which restricts their widespread application [35]. Mathematical models, including logistic functions, Savitzky–Golay filters, and wavelet transforms, are used to simulate time-series data of VIs [33], [36], [37], [38], with dates exhibiting specific mathematical characteristics in the fit curves that are defined as particular phenological events [39], [40]. For example, the four main phenological stages (e.g., the start of the season, maturity, senescence, and the end of the season) are identified at critical points in the fit VI time-series using piecewise logistic models [38]. However, these inflection points in the fit time series do not always align with the phenological periods of crops [41], [42]. SMF methods use geometric pattern matching to align VI time-series with field-observed crop growth stages. This eliminates reliance on predefined mathematical functions, demonstrating greater robustness and adaptability [43]. Compared with traditional methods, SMF approaches can integrate ground phenological observations, linking the phenological stages of specific crops directly to VI time series. This significantly enhances the spatiotemporal transferability of models [44]. This advantage highlights their potential for addressing issues arising from phenological mismatches when estimating winter wheat parameters using remote sensing.

Integrating phenological data are crucial for enhancing the accuracy and applicability of crop estimation [28], [29], [45]. Pei et al. [46] compared the effectiveness of ground-based and remote sensing phenology in predicting crop yields. They found that subdividing the time window by phenological stages more accurately extracts environmental variables and captures the complex interaction between crop development and external factors. This improves the accuracy of yield estimation. Zhang et al. [47] used satellite-based phenology in deep learning models to improve the estimation of large-scale crop yields. These studies all demonstrate the importance of refining phenological periods and the impact of phenological alignment on model accuracy. However, data-driven models are still influenced by the datasets used, and complex models are difficult to apply on a large scale. The crop biomass assessment for wheat (CBA-Wheat) model, proposed by Li et al. [28], is a relatively simple phenology-driven model that can estimate the continuous changes of AGB in winter wheat. In addition, Yang et al. [45] investigated the continuous relationship between VIs and phenology, developing a model applicable to different years. Zhao et al. [29] extended the research to encompass the estimation of multiple winter wheat parameters. Since the remote sensing technology can accurately obtain phenological information over a large area [27], this is a valuable tool for such studies. Integrating large-scale phenological information into phenology-driven models enables dynamic estimation at a regional scale.

The transferability of models from UAV to satellite mainly depends on their ability to adapt to various temporal and spatial dimensions. Existing studies have improved the model transferability by combining remote sensing and crop models, transfer learning algorithms, and phenological information. However, crop models often require extensive field-measured data and complex parameter calibration, which makes them difficult to transfer across different years and ecological sites [48]. Although transfer learning algorithms reduce reliance on labeled data to some extent, discrepancies in features between the source and target domains still limit the performance of model transfer [49]. In contrast, methods for acquiring phenological information are relatively simple and well-established, offering the potential to construct efficient, highly transferable estimation models. Integrating prior knowledge into the model transfer process reduces the effect of variations in data scale on model prediction accuracy, allowing further refinement of model parameters to enhance applicability [50], [51], [52], [53]. This technical approach provides robust support for the efficient transfer of models between UAV and satellite data.

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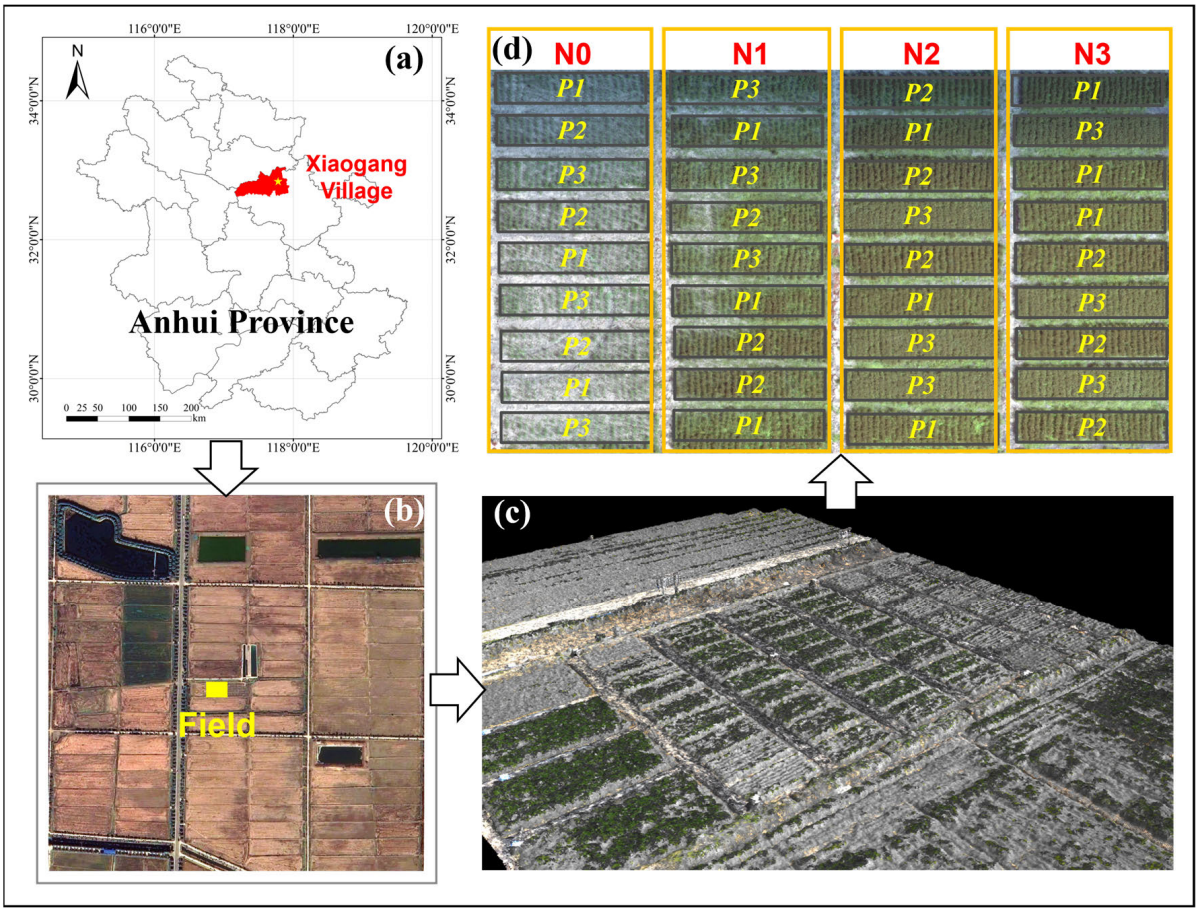


Fig. 1. Experimental design of winter wheat. (a) Geographical location. (b) Aerial image. (c) 3-D real scene photograph of the experiment. (d) Experimental planting diagram. (Note: N is the nitrogen fertilizer level, and P is the winter wheat variety).

Although many current studies have demonstrated the importance of data transfer and phenological periods in crop estimation, research combining phenological matching and data transfer is lacking. The key objectives of this study are: 1) assessing the influence of various model transfer strategies on the accuracy of cross-platform applications; 2) exploring the effectiveness of phenological matching techniques in enhancing the transfer of phenology-driven models from UAV to satellite platforms; and 3) achieving high-precision estimation on satellite platforms using models initially constructed with UAV data through a framework that integrates phenological calibration and data transfer.

II. Materials and Methods

A. Experimental Design

As shown in Fig. 1, field experiments on winter wheat were conducted in Xiaogang Village, Fengyang, Chuzhou, Anhui, from 2021 to 2023.

In 2021, a total of 36 plots were cultivated, each measuring 2×8 m. Three winter wheat cultivars were used: Huaimai 44 (P1), Yannong 999 (P2), and Ningmai 13 (P3). These were subjected to four nitrogen (N) fertilizer treatments and three replicates. Sowing took place on November 25, 2020, and harvesting occurred in early June 2021. Prior to sowing, potassium fertilizer (potassium chloride with a K_2O content of 50%) and phosphorus fertilizer (calcium superphosphate with a P_2O_5 content of 12%) were applied as base fertilizers at rates of 135 kg/hm^2 (in terms of K_2O, pure) and 90 kg/hm^2 (in terms of P_2O_5, pure), respectively. Granular urea nitrogen fertilizer (46% N) was applied in two splits: 60% as a basal application and 40% as a top dressing during the jointing stage. Four nitrogen treatments were applied: N0 (no nitrogen), N1 (100 kg/hm^2), N2 (200 kg/hm^2), and N3 (300 kg/hm^2). No biotic or abiotic stresses occurred during the experiment, and local high-yield farming practices were followed for field management.

The design in 2022 was consistent with that in 2021. Sowing took place on November 19, 2021, with plot dimensions of 2×5 m, and fertilization and management practices remained unchanged from the previous year.

In 2023, the design remained consistent with that of 2021. Sowing took place on November 10, 2022, with plot dimensions of 2×5 m, and the cultivars used were Huaimai 51 (P1), Yannong 999 (P2), and Ningmai 13 (P3). Fertilization and management practices stayed the same as in 2021.

B. Data Acquisition and Preprocessing

1) Field UAV Data Acquisition: This study used DJI Phantom 4 Multispectral (P4M) and DJI Mavic 3 Multispectral (M3M) drones (both DJI Technology, Shenzhen, China) to

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TABLE I
UAV IMAGE AND AGB ACQUISITION DATE

Year	The date of image acquisition and the corresponding growth stage	UAV platform
2021	Jointing (03/14), booting (04/08), heading (04/18), anthesis (04/29), filling (05/09), mature (05/24)	P4M
2022	late tillering (02/24), jointing (03/16), booting (04/10), filling (05/05), mature (05/21)	
2023	late tillering (02/25), jointing (03/14), booting (04/08), anthesis (04/19), filling (05/08), mature (05/19)	M3M

TABLE II
STATISTICAL ANALYSIS OF THE AGB OF WINTER WHEAT IN THE FIELD OVER THREE YEARS

Year	Number	Mean (kg/hm^2)	Max (kg/hm^2)	Min (kg/hm^2)	Standard deviation	Coefficient of variation
2021	216	8406.83	24481.83	391.11	5788.96	0.69
2022	180	9430.40	25059.19	453.56	7953.21	0.84
2023	216	9148.93	30956.58	871.55	7118.22	0.79

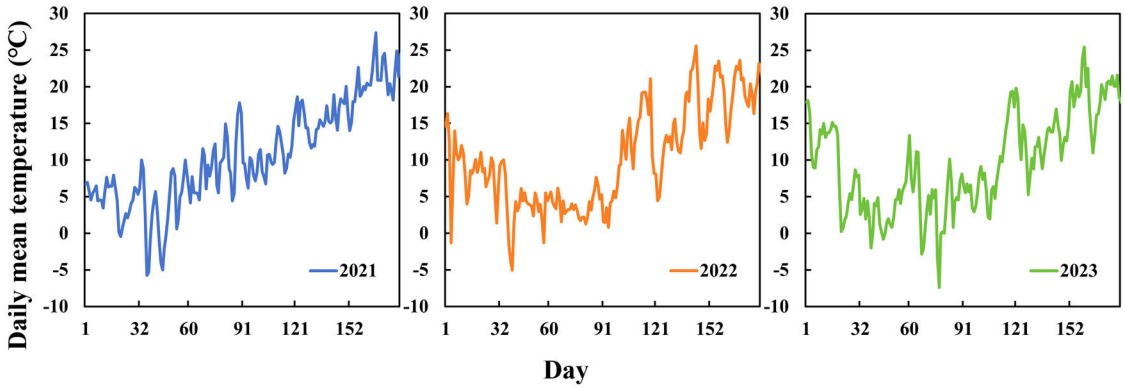


Fig. 2. Changes in the daily mean temperature in the study area over three years (from the winter wheat planting date to the harvest date).

capture multispectral imagery of winter wheat. The P4M is equipped with a visible light imaging sensor and five multispectral sensors: blue: 450 ± 16 nm, green: 560 ± 16 nm, red: 650 ± 16 nm, red edge: 730 ± 16 nm, and near infrared: 840 ± 26 nm). The M3M features a visible light sensor and four multispectral imaging sensors. Compared with the P4M, it lacks the blue band. Images for 2021 and 2022 were captured using the P4M, while the M3M was used for imagery in 2023.

The date of image acquisition is detailed in Table I. The UAV flight setup and imagery processing workflow can be found by Zhu et al. [7].

2) Field AGB Data Acquisition: The AGB of winter wheat was collected on the same day as the UAV operations. Details of the acquisition and processing of the winter wheat AGB sample can be found in Zhu et al. [7]. The results of the statistical analysis of the three-year field AGB measurements are shown in Table II.

3) Field Scale Meteorological Data and Different Phenological Calculation Methods: Meteorological data were collected throughout the three-year field study of winter wheat (see Fig. 2). The average daily temperature, derived from the daily maximum and minimum values, is used to calculate heat-related phenological methods. In addition, eight digital growth stage methods have been defined (see Table III).

TABLE III
DIFFERENT METHODS OF DIGITIZED GROWTH STAGES

Method	Formula
DOY_0101 DOY_sow	$DOY = \sum_{1}^{t} DOY$
AGDD_0101 AGDD_sow	$AGDD = \sum_{1}^{t} \frac{(T_{max} - T_{min})}{2} - T_{base}$
RGS_DOY_0101 RGS_DOY_sow	$RGS_DOY = \frac{DOY_t}{DOY_{max}}$
RGS_AGDD_0101 RGS_AGDD_sow	$RGS_AGDD = \frac{AGDD_t}{AGDD_{max}}$

Note: In the formula, T_{max} and T_{min} represent the highest and lowest daily temperatures, respectively. T_{base} is the base temperature for winter wheat growth, which is set at 0°C in this chapter. t refers to the number of days or accumulated growing degree days (AGDD) at the time of image acquisition. max refers to the maximum number of days or maximum AGDD. "0101" indicates that the time calculation starts from January 1st each year, while "sow" represents the time calculation starting from the sowing date.

4) Satellite Image Data: The Moderate Resolution Imaging Spectroradiometer (MODIS) instruments on the Terra and Aqua satellites provide reliable data for the creation of associated products by delivering near-daily global coverage

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TABLE IV
STATISTICAL ANALYSIS OF SATELLITE-SCALE AGB IN 2010

Year	Number	Mean (kg/hm^2)	Max (kg/hm^2)	Min (kg/hm^2)	Standard deviation	Coefficient of variation
2010	21498	4020.85	19288	25	4539.91	1.12

TABLE V
VIS USED IN THIS STUDY

Vegetation index	Formula	Reference
CIgreen	$\frac{R_{nir}}{R_{green}} - 1$	[54]
NDVI	$\frac{R_{nir} - R_{red}}{R_{nir} + R_{red}}$	[55]
NIRv	$R_{nir} \times \frac{R_{nir} - R_{red}}{R_{nir} + R_{red}}$	[56]
WDRVI	$\frac{0.1 \times R_{nir} - R_{\text{red}}}{0.1 \times R_{nir} + R_{red}}$	[57]
PSSRa	$\frac{R_{nir}}{R_{red}}$	[58]
SAVI	$1.5 \times \frac{R_{\text{nir}} - R_{red}}{R_{\text{nir}} + R_{red} + 0.5}$	[59]
EVI2	$2.5 \times \frac{R_{nir} - R_{red}}{R_{nir} + 2.4 \times R_{red} + 1}$	[60]

Note: R_g refers to the reflectance of the green band; R_{red} denotes the reflectance of the red band; R_{nir} indicates the reflectance of the near-infrared band.

of land surfaces. For this study, we selected the MOD09A1 product, which provides composite surface reflectance data at a resolution of 500 m, updated every eight days. These data meet the requirements for the large-scale, high-frequency estimation of AGB of winter wheat.

We utilized the Google Earth Engine (GEE) platform to acquire the satellite data from January 1 to June 29 for each year between 2007 and 2015, and conducted a series of preprocessing operations, including cloud removal. To ensure consistency with the spectral bands obtained by UAV, this study extracted reflectance data from three specific bands: sur_refl_b01 (620–670 nm), sur_refl_b02 (841–876 nm), and sur_refl_b04 (545–565 nm). These data will be used for subsequent model transfer and validation from UAV to satellite, with the aim of achieving seamless integration and application of models across different platforms.

5) Satellite Scale AGB Data: The aim of this study is to assess the ability of the model to transfer from the UAV scale to the satellite scale. For this purpose, we used the winter wheat AGB dataset for China from 2007 to 2015, which was developed by Huang et al. [3]. This dataset was created by combining the MODIS leaf area index (LAI) product with the WOFOST model assimilation technique. It covers 11 key wheat-producing regions in China, including Henan, Shandong, Anhui, Jiangsu, Hebei, Hubei, Shanxi, Shaanxi, Sichuan, Xinjiang, and Gansu. The dataset provides daily winter wheat AGB dynamics at a 1-km resolution from 2007 to 2015, offering valuable baseline data for our research. When validated against ground measurement data, the dataset demonstrated high consistency between its estimated AGB and field observations, with a correlation coefficient (R^2) exceeding 0.9 and a root-mean-square error (RMSE) of only 1377 kg/hm^2. Further comparison with county-level statistics from 2007 to 2015 revealed an R^2 ranging from 0.73 to 0.89, an RMSE ranging from 953 to 1503 kg/hm^2, and a mean absolute percentage error (MAPE) ranging from 8% to 12%. This indicates high accuracy and reliability at the regional scale. Based on the imaging dates of the MOD09A1 product, we precisely matched and extracted the AGB data corresponding to the same dates from the winter wheat AGB dataset in China for 2007–2015, thereby ensuring the data's temporal and spatial consistency. On this basis, for each aligned date, 1000 pixels were randomly selected to construct the satellite-scale validation dataset. However, due to fluctuations in the number of pixels in the winter wheat AGB dataset on different dates, as well as the absence of some pixels in the satellite data, the validation dataset for certain dates did not meet the 1000-pixel threshold. Table IV shows the statistical analysis of the satellite-scale AGB data from January 1, 2010 to June 29, 2010.

6) VI Calculation: Due to the differences in spectral band configurations across the various datasets of imagery used in this study, we selected the green, red, and near-infrared bands to calculate VIs, thereby ensuring consistency and comparability in the analysis. The specific calculation formulas are provided in Table V.

III. WORKFLOW

Fig. 3 shows the workflow of this research, which is divided into two main sections. The first section uses UAV-scale data to investigate methods for model transfer and application. The second section applies the models developed at the

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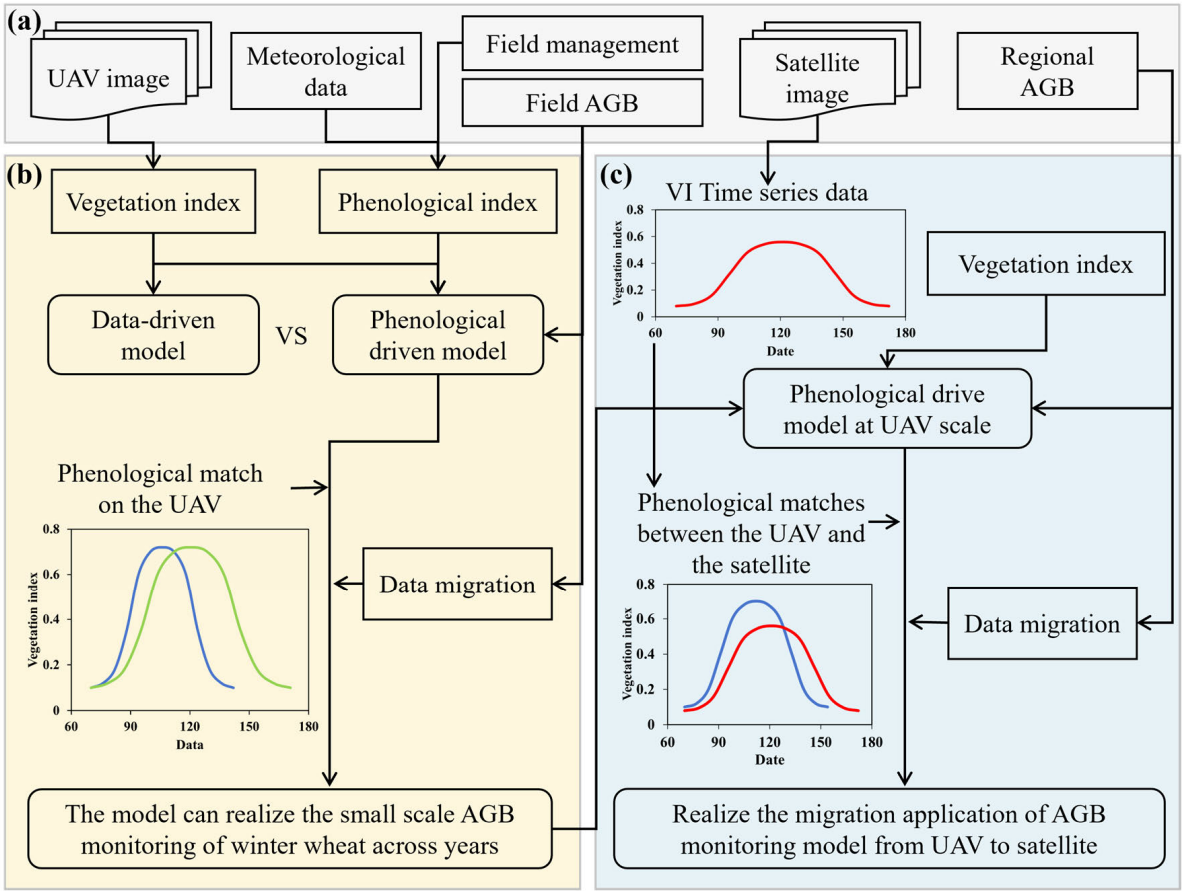


Fig. 3. Flowchart outlines the main operations of this study. (a) UAV and satellite images, along with their corresponding AGB data. (b) Verify the feasibility of the methodology at the UAV scale. (c) Transferring the UAV-scale model to the satellite scale.

UAV scale to the satellite scale, using methods established in previous studies. The main data sources are UAV imagery, field-measured winter wheat AGB data, field meteorological data, satellite imagery, and satellite-scale AGB data.

First, the study employed UAV multispectral imagery and field-measured winter wheat AGB data to assess the accuracy of various AGB estimation models. It also examined the impact of phenological matching and data transfer strategies on the implementation and transfer of phenology-driven models at the UAV scale. Subsequently, the phenology-driven models developed at the UAV scale were successfully transferred to the satellite scale by employing phenological matching and data transfer strategies, thereby achieving the model transfer and application from the UAV to the satellite scale.

A. Construction of Phenology-Driven Model

This study developed three types of AGB estimation models. The first is a traditional linear model; the second is a random forest model; and the third is a phenology-driven model. Building on the phenology-driven model proposed by Li et al. [28], this study aims to address the issue of discontinuity in traditional models for estimating AGB of winter wheat. First, we calculate the univariate linear equations between VIs and AGB for different growth stages (1) to obtain the slopes (k) and intercepts (b) of these equations for each stage. We then fit the relationships between k and b, as well as the phenological periods (2). Finally, we constructed a continuous phenology-driven model (3)

$$\mathrm{AGB} = k \times \mathrm{VI} + b \tag{1}$$

$$k = \mathrm{Fun}(X), \quad b = \mathrm{Fun}(X) \tag{2}$$

$$\mathrm{AGB} = \mathrm{Fun}(X)_k \times \mathrm{VI} + \mathrm{Fun}(X)_b \tag{3}$$

where k and b represent the slope and intercept, respectively, of the linear model between VIs and AGB at different growth stages. X denotes the phenological periods, and Fun represents the fitting function.

B. UAV Scale

1) Influence of Different Data Transfer and Modeling Strategies on Model Transferability: To investigate the impact of potential factors on model transferability, such as different time window transfers and model construction processes, this study established ten strategies (see Fig. 4). The differences among these strategies mainly lie in data transfer at different time windows (see Table VI). In 2023, each growth stage corresponds to a single period. The 2023 stage periods are combinations of two adjacent growth stages (stage 1: February 25, 2023 and March 14, 2023; stage 2: April 8, 2023 and April

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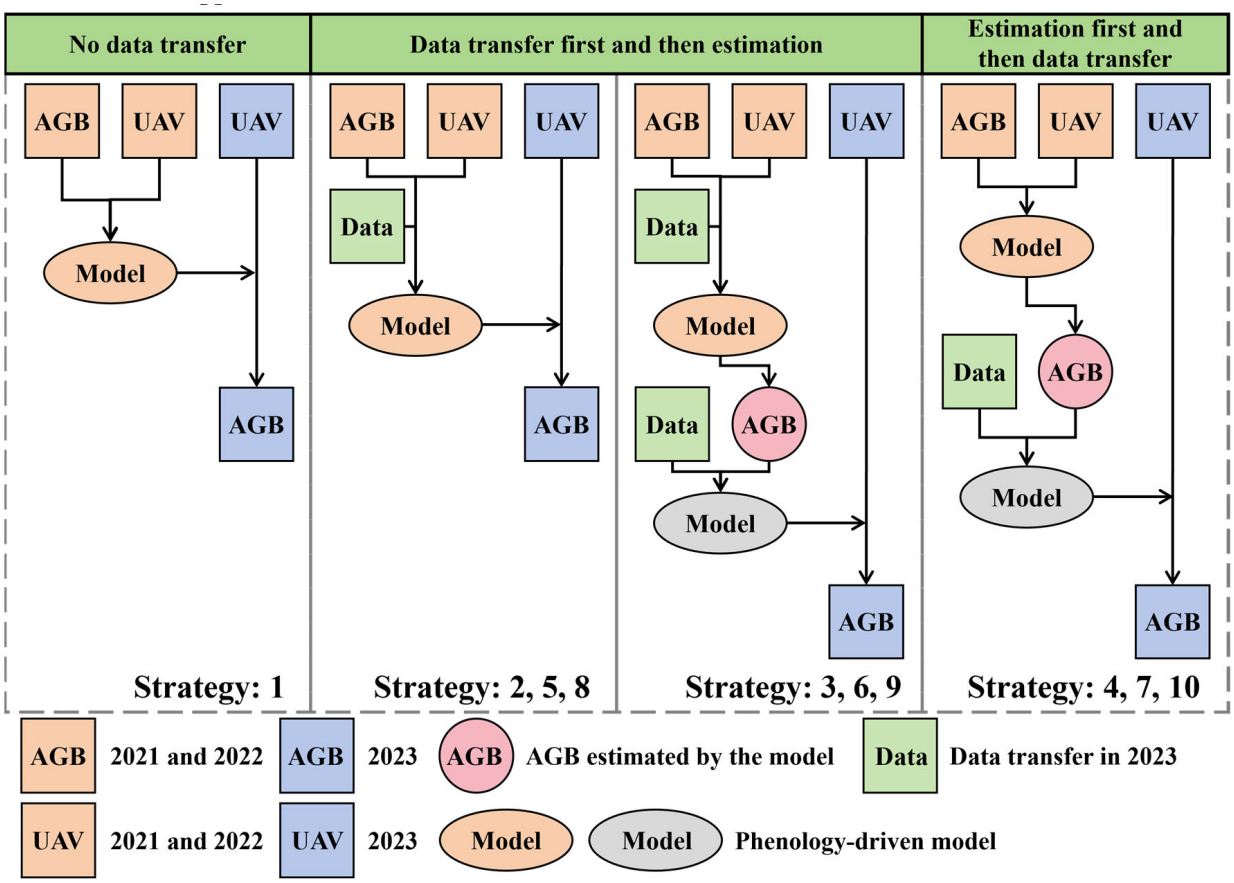


Fig. 4. Schematics of different strategies. (Note: Under the same process, the differences among strategies mainly lie in data transfer at different time windows.)

TABLE VI
TIME WINDOW INVOLVED IN EACH STRATEGY

Time window	Description	Strategy
None	No data transfer from 2023	1
Single period	Each individual growth stage in 2023	2, 3, 4
Stage period	Two adjacent growth stages in 2023	5, 6, 7
Cumulative period	The first few growth stages in 2023	8, 9, 10

19, 2023; and stage 3: August 5, 2023 and May 19, 2023). The cumulative periods in 2023 are combinations of the preceding stages (e.g., cumulative period 1 includes the first growth stage of 2023 and cumulative period 3 includes the first three growth stages of 2023).

As shown in Fig. 4, the model construction process is divided into four situations. Consider strategy 6 for illustration purposes. The initial modeling set for strategy 6 includes all the data from 2021 to 2022. Then, the data from each stage in 2023 is transferred to the initial modeling set, and the phenology-driven model is constructed. Next, the AGB data estimated by the phenology-driven model is transferred with the 2023 data, and the second phenology-driven model is applied to all the 2023 data.

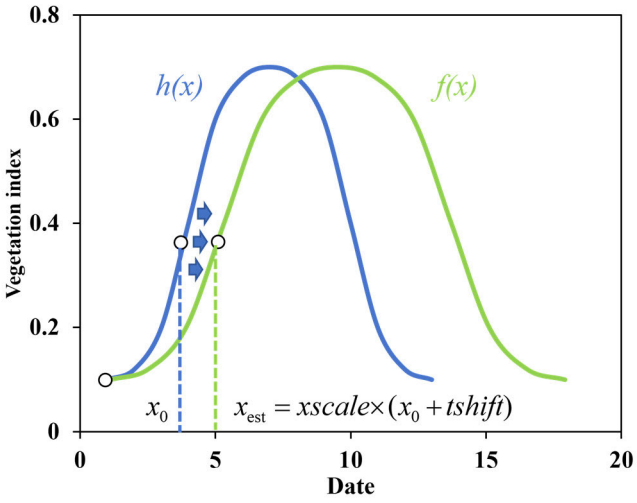


Fig. 5. Schematic of the phenological matching.

2) Impact of Phenological Matching on Model Transferability: As shown in Fig. 5, the time-series trajectories of VIs for winter wheat often vary considerably across years and locations. This is influenced by variations in environmental factors and management practices. This results in inconsistent phenological periods affecting the model when it is transferred

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and applied. Therefore, the phenological calibration is necessary to reduce errors caused by the phenological mismatch.

To achieve this objective, this study adopted the SMF method, which was proposed by Sakamoto et al. [43] and has been shown to be highly effective in identifying phenological information for a variety of crops.

1) First, build the shape model $h(x)$ using the time-series data of VIs from the modeling dataset, as illustrated by the blue line in Fig. 5.
2) Next, fit a curve $f(x)$ to the VI time-series from the validation dataset, as shown by the green line in Fig. 5.
3) Then, identify the optimal parameters ($x$scale, $y$scale, and $t$shift) between $h(x)$ and $f(x)$ through linear shifting and scaling steps (4). The optimal scaling parameters are determined based on the minimum RMSE between $g(x)$ and $f(x)$ (5).
4) All phenological stages (x_{est}) in $f(x)$ can be derived from the corresponding stages (x_0) in $h(x)$ using $x$scale and $t$shift (6) and (7).
5) Finally, reconstruct the phenology-driven model using the data after phenological matching to enable the model to be transferred and applied to the validation dataset

$$g(x) = y\text{scale} \times h(x\text{scale} \times (x_0 + t\text{shift})) \tag{4}$$

$$\text{RMSE} = \left[\frac{1}{n}\left(\sum((f(x) - g(x)))^2\right)\right]^{\frac{1}{2}} \tag{5}$$

$$x_{\text{est}} = x\text{scale} \times (x_0 + t\text{shift}). \tag{6}$$

where $x$scale and $y$scale are the scaling factors in two dimensions (i.e., time and VI value), while $t$shift is the time offset factor.

This study used the particle swarm optimization (PSO) algorithm to find the optimal solutions for these three parameters. The PSO algorithm was configured with a swarm size of 50 particles and a maximum of 200 iterations. The ranges for the three factors were set as follows: $0.3 < x\text{scale} < 3$, $0.5 < y\text{scale} < 3$, and $-100 < t\text{shift} < 100$.

Based on the work of Liu et al. [34], an error was identified in (6), and their paper provides a detailed explanation of this issue. In practical applications, we used the following equation to calculate x_{est} after phenological matching:

$$x_{\text{est}} = \frac{1}{x\text{scale}} \times x_0 - t\text{shift}. \tag{7}$$

We will calculate the degree of fit between the two VI curves using the optimal parameters of the PSO search

$$R^2_\text{best} = 1 - \frac{\sum(f(x_{\text{est}}) - h(x_{\text{est}}))^2}{\sum\left(h(x_{\text{est}}) - \overline{h(x_{\text{est}})}\right)^2}. \tag{8}$$

C. Transfer and Application of Model From the UAV Scale to the Satellite Scale

Building on the results of UAV-scale models under various strategies, as well as the influence of phenological matching on model performance, this study applied UAV-scale models directly to the satellite scale. This was to explore the potential of phenology-driven models across different locations and times, as well as the reliability of transfer strategies. There are three main forms of transfer.

1) *Direct Transfer:* The phenology-driven model, constructed using UAV data and field-measured AGB data from 2020 to 2021, was applied directly to the satellite scale for the years 2007–2015, without any phenological matching or data transfer.
2) *Phenological Matching:* Before transferring the UAV model to the satellite scale, the phenological stages of the UAV model were matched to those at the satellite scale using the time-series curves of VIs from both UAV and satellite data. The model was then applied after this matching process.
3) *Phenological Matching + Data Transfer:* Building on the previous phenological matching, this step investigated the impact of data transfer under strategy 3 for both single and cumulative periods. Although eight digital phenological methods were previously mentioned, obtaining sowing dates for winter wheat at different times and in different locations is challenging. Furthermore, to ensure the universality and simplicity of model transfer, the UAV to satellite transfer relied solely on the date of year (DOY)_0101. Six dates with corresponding UAV and satellite observation times were selected for analysis in order to evaluate the effectiveness of the model transfer process.

D. Statistical Analysis

The accuracy of estimating photosynthetic parameters from remote sensing data was assessed using the coefficient of determination (R^2), RMSE, and relative RMSE (nRMSE), defined as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - x_i)^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2} \tag{9}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - x_i)^2} \tag{10}$$

$$\text{nRMSE} = \frac{\text{RMSE}}{\overline{x}} \tag{11}$$

where y_i and x_i represent the predicted and measured winter wheat AGB for sample i, respectively, $\overline{x}$ represents the average measured winter wheat AGB, and n is the total number of samples.

IV. Results

A. Constructing AGB Estimation Models at the UAV Scale

1) Impact of Phenological Stages on the Data-Driven Model: Data-driven approaches utilizing VIs are commonly employed to assess the crop physiological and biochemical parameters due to their simplicity. However, the estimation of winter wheat AGB using the empirical model approaches can be affected by phenological effects, and the discontinuity of the models also limits their transferability and application. Fig. 6 uses the wide dynamic range VI (WDRVI) under the DOY_0101 index in 2021 as an example to investigate the impact of phenology on traditional regression methods.

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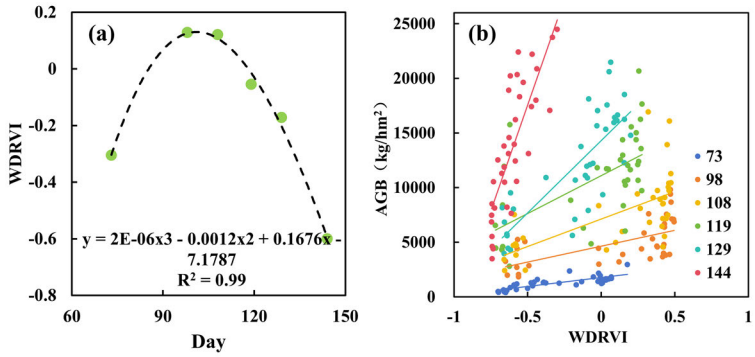


Fig. 6. Influence of the phenological effects on the relationship between WDRVI and AGB. (a) Changing trend of WDRVI over time. (b) Relationship between AGB and WDRVI at different phenological periods.

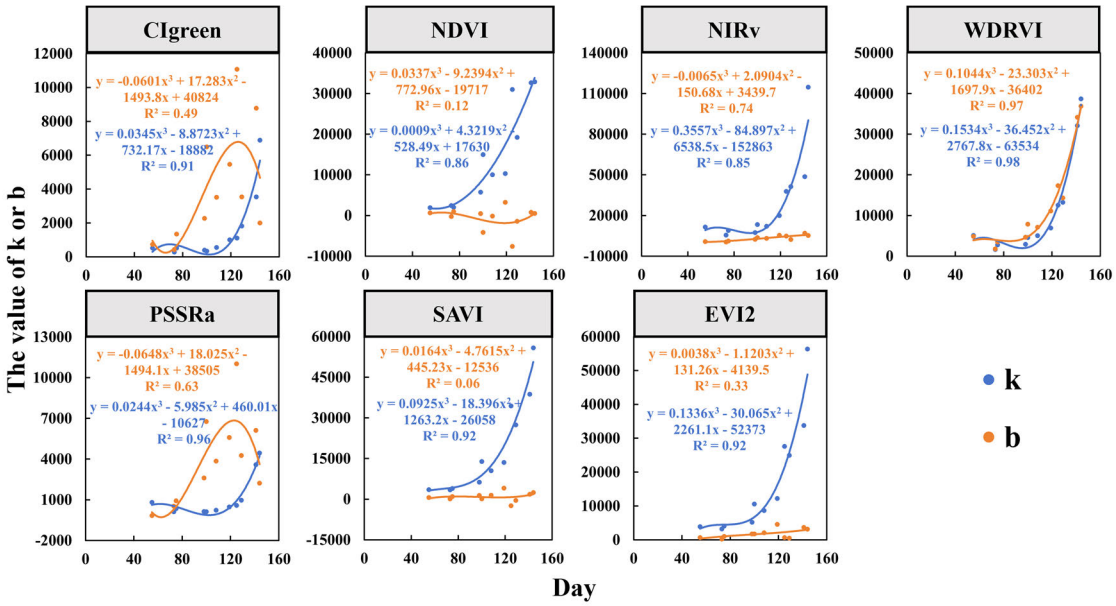


Fig. 7. Changes of coefficient values slope (k) and intercept (b) in the AGB Linear model with the DOY_0101 for the growth stage of winter wheat.

Fig. 6(a) illustrates the trend of WDRVI changing across different phenological stages, revealing a significant shift in the relationship between WDRVI and AGB around the heading stage (DOY_0101 = 108). Empirical model approaches construct estimation models by assuming a one-to-one mapping relationship between VI and AGB. However, multiple AGB values correspond to a WDRVI value of 0 [see Fig. 6(b)], which makes the relationship between VIs and AGB highly susceptible to phenological stages. Discontinuous traditional regression models can make it difficult to select an appropriate estimation model between two phenological stages. As illustrated by the black dashed line in Fig. 6(b), selecting one of the two models between the filling stage (DOY_0101 = 129) and the heading stage (DOY_0101 = 144) would result in a significant difference in estimated AGB. Therefore, it is of great significance to consider the impact of phenological stages on models.

2) Linear Relationship Between VIs and AGB at Different Growth Stages: Fig. 7 uses data from 2021 to 2022 to illustrate the linear relationships between seven VIs and AGB at various growth stages, based on the DOY 0101. It also illustrates variations in the slope (k) and the intercept (b) at different stages. The results show that the k values of all indices increase over time, while the b values differ significantly between the indices. Specifically, the b value of WDRVI increases over time, showing the best fit with the phenological period with an R^2 value reaching 0.97. In contrast, the relationships between the b values of the other VIs and the phenological period generally show moderate alignment. Therefore, the impact of different strategies on model transferability is analyzed using the WDRVI index as an example.

3) Comparison of Phenology-Driven Model With Data-Driven Model: Using the DOY_0101 WDRVI indicator as an example, AGB estimation models were constructed using data from 2021 to 2022 and then validated using data from 2023. The performance outcomes of the three AGB models are provided in Table VII. The phenology-driven model significantly outperforms the linear regression and random forest models. The phenology-driven model predicts AGB in 2023 (R^2 = 0.78 and nRMSE = 0.37). A comparison of the

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TABLE VII
RESULTS OF ESTIMATING WINTER WHEAT AGB USING PHENOLOGY-DRIVEN MODEL AND DATA-DRIVEN MODEL

Year	2021, 2022		2023	
	R^2	nRMSE	R^2	nRMSE
Phenology-driven model	0.79	0.35	0.78	0.37
Linear regression model	0.09	0.74	0.08	0.75
RF model (single VI)	0.86	0.29	0	0.95
RF model (all VIs)	0.94	0.19	0	0.91

TABLE VIII
MODEL TRANSFER PERFORMANCE UNDER DIFFERENT STRATEGIES

Strategy	Time window	Iteration	R^2	nRMSE
1	No data transfer	0	0.84	0.33
2		1	0.84	0.33
3	Single period	2	0.84	0.32
4		1	0.84	0.33
5		1	0.84	0.32
6	Stage period	2	0.85	0.32
7		1	0.84	0.32
8		1	0.85	0.32
9	Cumulative period	2	0.85	0.31
10		1	0.85	0.32

phenology-driven model with the traditional regression model at different stages is presented in Tables SI and SII of the Supplementary Material. The results also indicate that the phenology-driven model demonstrates higher accuracy than traditional data-driven models, whether based on the annual data or corresponding phenological stage data.

4) Comparison of Different Model Construction Strategies for Phenology-Driven Model: Table VIII presents the changes in model accuracy under different data transfer processes and modeling procedures, taking DOY_0101 as an example. The results show that models that use data transfer (strategies 2–10) are more accurate than models that do not use data transfer (strategy 1). Models that experience two rounds of data transfer and modeling processes demonstrate superior transfer performance under the same data transfer treatment (strategies 3, 6, and 9). Among the strategies, strategy 9 achieves the best performance, with an R^2 of 0.85 and an nRMSE of 0.31. Comparisons of models under different time window transfers are presented in Table SIII of the Supplementary Material. The results show that, for a single time window, models involving data transfer from the filling stage are more accurate. During the cumulative period, the greater the number of data transfer stages, the more suitable the AGB estimation model becomes for the target dataset.

5) Impact of Phenological Matching on Model Accuracy: For example, using DOY_0101, a cubic polynomial function was employed to model the relationship between time and VIs, utilizing the observed correlation in both the modeling dataset (2021 and 2022) and the validation dataset (2023). Phenological matching was then performed using a shape-fitting approach. The results for the seven VIs before and after phenological matching are shown in Table IX, where R^2_best represents the fitting accuracy. Most VIs achieved improved accuracy after phenological matching, except for WDRVI, which experienced a slight decline in precision. This study also investigated the impact of phenological matching on various VIs, with the results presented in Table SIV of the Supplementary Material. It was found that most VIs exhibited enhanced accuracy following phenological matching.

6) AGB Mapping at the UAV Scale: The AGB of winter wheat in 2023 was estimated using the phenology-driven model, as illustrated in Fig. 8. Variability in AGB was evident both within and between plots. Variations in AGB across different nitrogen fertilizer treatments were clearly distinguishable, and the temporal patterns of AGB throughout the full-growth cycle of winter wheat were accurately captured. These findings provide valuable insights for the precise management of winter wheat.

B. Application of Transferring AGB Estimation Model From UAV to Satellite

1) Impact of Direct Transfer and Phenological Matching on Phenology-Driven Model: Using the 2010 satellite data as an example, the phenology-driven, linear, and random forest models at the UAV scale were transferred directly to the satellite scale. The results of the three models at the satellite scale are shown in Table X. These results demonstrate that none of the three models can be transferred directly from the UAV scale to the satellite scale for AGB estimation.

To improve the accuracy of the models at the satellite scale, we applied the phenological matching technique, which performed well at the UAV level, to align the phenological stages between the UAV and satellite data. The results are presented in Table XI. These results show that, except for the WDRVI and PSSRa indices, which cannot perform good phenological matching, the accuracy of AGB estimation for the other indices is significantly improved after phenological matching. The soil-adjusted VI (SAVI) model showed the most significant improvement in accuracy among all the models, with an R^2 of 0.68 and an RMSE of 3530.92 kg/hm^2. The results of the transfer from UAV to satellite in other years are shown in Table SV of the Supplementary Material.

2) Impact of Data Transfer and Phenological Matching on the Transfer and Application of Phenology-Driven Model from UAV to Satellite: Research results at the UAV scale indicate that, for a single-period data transfer, stages closer to later growth stages (when AGB values are higher) have a more significant impact on the improvement in the accuracy of phenology-driven models. For cumulative period data transfer, the greater the number of stages included in the data transfer, the greater the model accuracy. Of the various strategies examined, conducting two rounds of modeling was found to significantly enhance model accuracy. Consequently, this study examines the influence of single and cumulative periods on the performance of phenology-driven models in the context of two modeling rounds. Taking SAVI in 2010 as an example, the results are presented in Table XII. The analysis shows that data from stages near the later growth phases have a stronger impact on the model, which is consistent with observations at the UAV scale. Combining data transfer with

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TABLE IX

INFLUENCE OF DIRECT TRANSFER AND PHENOLOGICAL MATCHING ON THE ACCURACY OF PHENOLOGY-DRIVEN MODEL

VI	Direct transfer				Phenological matching				R^2_best
	Train		Test		Train		Test		
	R^2	nRMSE	R^2	nRMSE	R^2	nRMSE	R^2	nRMSE	
CIgreen	0.78	0.36	0.76	0.38	0.78	0.36	0.80	0.35	0.76
NDVI	0.82	0.33	0.83	0.32	0.82	0.33	0.86	0.29	0.27
NIRv	0.75	0.38	0.82	0.33	0.75	0.38	0.86	0.29	0.96
WDRVI	0.79	0.35	0.78	0.37	0.79	0.35	0.74	0.39	0.98
PSSRa	0.63	0.47	0.60	0.49	0.63	0.47	0.62	0.48	0.94
SAVI	0.81	0.34	0.84	0.31	0.81	0.34	0.88	0.27	0.90
EVI2	0.80	0.34	0.84	0.31	0.80	0.34	0.88	0.27	0.93

TABLE X

ACCURACY OF THE THREE AGB MODELS DIRECTLY TRANSFERRED FROM THE UAV SCALE TO THE SATELLITE SCALE

VI	Phenology-driven model		Linear regression model		Random forest model	
	R^2	RMSE (kg/hm^2)	R^2	RMSE (kg/hm^2)	R^2	RMSE (kg/hm^2)
CIgreen	0	10181.96	0	5393.62	0	7343.86
NDVI	0	9454.30	0	4916.39	0	6989.40
NIRv	0	9388.91	0	5097.25	0	6825.24
WDRVI	0	8361.33	0	5336.61	0	6987.03
PSSRa	0	10241.50	0	5913.10	0	6988.29
SAVI	0	10295.73	0	4940.27	0	6704.32
EVI2	0	9712.18	0	4993.24	0	7027.97

TABLE XI

INFLUENCE OF DIRECT TRANSFER AND PHENOLOGICAL MATCHING ON THE ACCURACY OF MODEL TRANSFER FROM UAV SCALE TO SATELLITE SCALE

VI	Direct transfer		Phenological matching		R^2_best
	R^2	RMSE (kg/hm^2)	R^2	RMSE (kg/hm^2)	
CIgreen	0	10181.96	0.04	4453.36	0.75
NDVI	0	9454.30	0.51	3166.16	0.95
NIRv	0	9388.91	0.43	3439.70	0.95
WDRVI	0	8361.33	0	29992.59	0
PSSRa	0	10241.50	0	7058.20	0
SAVI	0	10295.73	0.68	2552.17	0.53
EVI2	0	9712.18	0.40	3530.92	0.98

TABLE XII

INFLUENCE OF DATA TRANSFER ON THE ACCURACY OF MODEL TRANSFER FROM UAV SCALE TO SATELLITE SCALE

Strategy	DOY_0101	Data transfer		Phenological matching + data transfer	
		R^2	RMSE (kg/hm^2)	R^2	RMSE (kg/hm^2)
Single period	57	0	11292.41	0.11	4274.75
	73	0	10299.95	0.66	2631.44
	97	0	9952.09	0.33	3720.75
	121	0	12331.40	0.24	3967.74
	129	0	9864.13	0.72	2417.62
	145	0	5157.75	0.75	2279.71
Cumulative period	57 ~ 73	0	11318.71	0.64	2712.56
	57 ~ 97	0	10870.84	0.20	4068.21
	57 ~ 121	0	13003.36	0.66	2635.29
	57 ~ 129	0	12228.91	0.72	2400.81
	57 ~ 145	0	6239.03	0.79	2086.09

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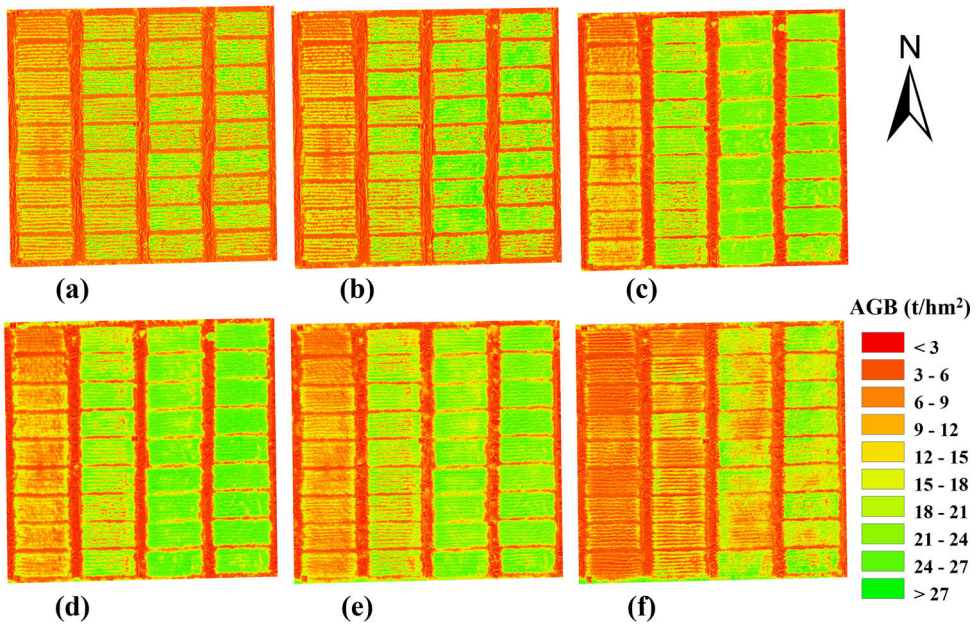


Fig. 8. Estimation mapping of AGB for winter wheat in 2023. (a) February 25, 2023. (b) March 14, 2023. (c) April 8, 2023. (d) April 19, 2023. (e) May 8, 2023. (f) May 19, 2023.

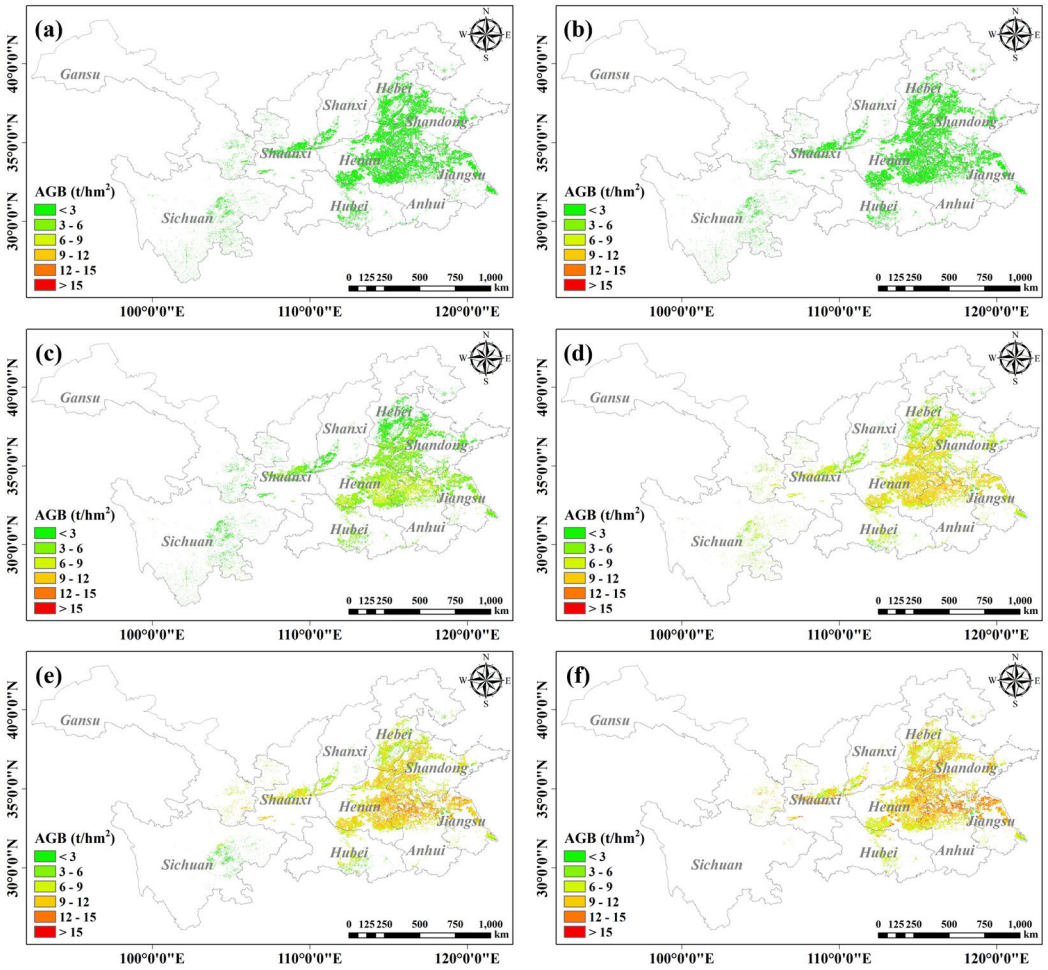


Fig. 9. AGB mapping of winter wheat for six dates in 2010 was estimated using the PDBM method. Each subplot represents the AGB estimation results for different dates. (a) February 26, 2010. (b) March 14, 2010. (c) April 7, 2010. (d) May 1, 2010. (e) September 05, 2010. (f) May 25, 2010.

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phenological matching significantly enhances model accuracy. Of the individual periods, days 129 and 145 have a more significant impact on the model. Day 145 shows the highest data transfer performance, reaching an R^2 of 0.75 and an RMSE of 2279.71 kg/hm^2. For cumulative periods, the highest accuracy is achieved when all stages are included, with an R^2 of 0.79 and an RMSE of 2086.09 kg/hm^2.

3) AGB Mapping at the Satellite Scale: Fig. 9 shows the spatial distribution of winter wheat AGB on six specific dates in 2010. This was derived from the phenological matching with data transfer (PDBM) of the SAVI index through phenological matching and a 145-day data transfer process. These maps provide an in-depth view of the spatial variability of AGB at various points throughout the year, offering insight into the growth dynamics of winter wheat. The northwest region, particularly Xinjiang, where winter wheat cultivation occupies less than 0.5% of the total administrative area, has sparsely populated regions, which makes it difficult to accurately represent pixel data on the 1-km AGB maps. Therefore, during the mapping process, we decided not to include data from Xinjiang. This was done to ensure the clarity and accuracy of the maps and to avoid providing misleading information due to sparse data.

In general, the retrieved AGB data is concentrated primarily in Chinese provinces of Hebei, Shandong, Anhui, and Henan. Model predictions show a clear upward trend over time. Further analysis reveals that the difference in AGB between China's northern and southern regions becomes more pronounced as the season progresses. As illustrated in Fig. 9(e), winter wheat AGB values in Hebei Province generally fall within the range of 6–12 t/hm^2, whereas in the provinces of Henan and Anhui, the values are higher, typically ranging from 9 to 15 t/hm^2. The model's ability to capture spatiotemporal fluctuations in AGB is crucial for understanding the regional crop growth disparities and provides a scientific basis for regional crop management strategies.

V. Discussion

A. Advantages of the Phenology-Driven Model

Currently, AGB estimation models based on the spectral information primarily rely on data-driven empirical models [9]. Due to their simple structure and small number of parameters, these models have been widely used to estimate winter wheat AGB. In this study, linear models corresponding to various growth stages were developed based on the relationship between WDRVI and AGB at each phenological stage (see Table SI of the Supplementary Material). Although these models perform poorly when estimating year-round, they achieve good accuracy when applied to similar phenological stages. However, differences in variety, environment, and management practices cause growth variations in winter wheat, resulting in the empirical relationships between spectral information and AGB failing when applied across scales [28], [29], [45], [61]. This is because the connection between winter wheat's spectral information and AGB evolves dynamically as the crop grows and develops. Using year-round data to construct the VI-AGB relationship conflicts with the assumptions of empirical models, resulting in an inability to establish a one-to-one correspondence between spectral information and AGB (see Fig. 6). Therefore, the transferability of empirical models is limited, and they cannot fully meet the practical needs of AGB estimation.

The phenology-driven model used in this study successfully resolves two significant problems associated with traditional regression models: the inability to monitor AGB continuously and the disregard for the impact of phenological stages on the relationship between VIs and AGB. Since phenological periods change continuously and the actual spectral variations of crops are also continuous and stable, abrupt discontinuities are rarely observed [62], [63]. According to the equation of the phenology-driven model [(3)], the model can theoretically capture the evolving relationship between VI and AGB more effectively across various phenological stages. This provides a notable benefit for continuously and dynamically estimating winter wheat AGB. High-frequency VI and AGB data enable the fit functional curves $\mathrm{Fun}(X)_k$ and $\mathrm{Fun}(X)_b$ to capture subtle changes in the relationship between VI and AGB more precisely, particularly around the temporal inflection points of VI. This allows the dynamic relationship between VI and AGB to be reflected more sensitively. When predicting winter wheat AGB across the full-growth stage, the phenology-driven model substantially outperforms traditional regression models, with an increase in R^2 of 0.70 and a decrease in nRMSE of 0.38 (see Table VII).

Although the phenology-driven model offers significant advantages over traditional regression models, its results are still essentially determined by the dataset used. Comparing the range of winter wheat AGB at the field scale for 2021 and 2022 with regional scale data for 2010 (see Tables II and IV) shows that nitrogen fertilizer gradient treatments at the plot scale lead to a broader AGB range at the field scale than at the regional scale. Consequently, UAV-scale models constructed using 2021 and 2022 data overestimate winter wheat AGB at the regional scale. Consequently, although models established directly using UAV data cannot achieve high-precision transfer to the satellite scale, significant improvements in model accuracy are observed when data from later growth stages are transferred (see Table XII). This improvement is attributed to the substantial difference in data volume between the UAV and satellite scales. When satellite-scale data are transferred, the larger satellite dataset tends to overshadow the patterns in UAV-scale data, aligning the model's patterns more closely with satellite-scale results. Therefore, the importance of incorporating prior knowledge in model transfer and application cannot be overstated [50], [52]. Indeed, several studies have used active learning techniques to select representative samples, enabling robust models to be developed even with limited datasets [52]. However, active learning involves sampling and testing the entire dataset. If the full dataset is unavailable in advance, the constructed model will not be highly accurate [64], [65].

B. Importance of Phenological Matching

Phenology is closely associated with crop growth and development because it indicates the stages of their growth and the

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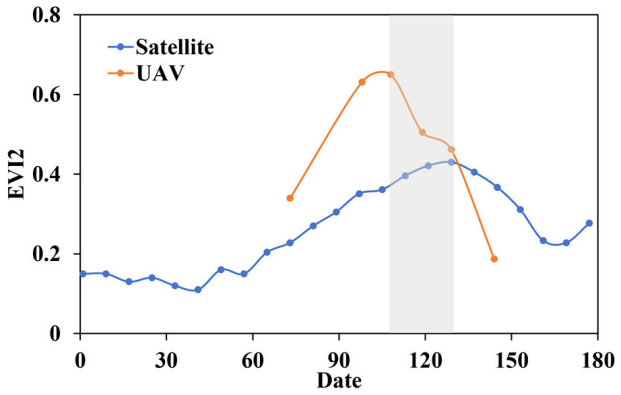


Fig. 10. Time-series curves of EVI2 at the UAV and satellite scales.

physiological changes they undergo in response to different environmental factors [66]. The phenological stages of winter wheat may also vary considerably under different $G \times E \times M$ conditions [67]. Fig. 10 shows the EVI2 time-series curves for UAV data in 2021 and satellite data in 2010. The UAV EVI2 curve shows a decline after day 108, while the satellite curve shows an increase during the same period. Direct application of the UAV-based phenology-driven model to satellite data would inevitably lead to erroneous AGB estimation due to the incorrect VI-AGB relationship at this time. Therefore, this study significantly improved the accuracy of model transfer from UAV to satellite through phenological matching (see Table XI), demonstrating that phenological matching between UAV and satellite data is crucial for model transfer and application.

Fig. S1 of the Supplementary Material shows the differences in model accuracy between the eight methods for digital growth stages at the UAV scale. The impact of the various methods on the model accuracy follows certain patterns, with relatively small differences in accuracy. Winter wheat grows and develops slowly throughout much of the winter, with substantial changes only occurring during the green-up stage, when its spectral curves start to exhibit noticeable variations [68]. At larger scales, the sowing dates and growth periods of winter wheat can vary significantly, making it difficult to determine accurate sowing dates across large areas [69]. Therefore, constructing time-series curves starting from January 1 each year simplifies the process of phenological matching.

Moreover, effective accumulated temperature is a key environmental factor that influences crop growth and development. Representing the heat accumulation necessary for crops throughout their growth, it significantly affects both the rate of development and the progression of phenological stages [70]. Calculation of the accumulated growing degree day (AGDD) through meteorological data is necessary. While many products (e.g., ERA5) are currently available to provide continuous temperature data on a global scale [71], it remains challenging to obtain accurate meteorological data on a small scale in agricultural systems, such as those in China, which are dominated by small-scale farming [72]. Research by Zhou et al. [73] has recently produced promising results in improving SMF for identifying phenological stages based on AGDD. Liu et al. [34] investigated the shape model fitting by separate phenological stages (SMF-S) method and found that AGDD is not suitable for estimating intraseasonal phenology. This is due to the AGDD model's use of a fixed base temperature of 3 °C. For winter wheat sown in the previous winter, the AGDD model may not accurately represent the early stages prior to spring green-up, as temperatures during this period often fall below the 3 °C threshold. Therefore, for phenological matching between UAV and satellite data, the method based on the DOY_0101 method is the simplest, most efficient, and most widely applicable.

The SMF method has two main issues. First, the target VI time-series curve is aligned by applying linear shifts and scaling to the shape model. However, this global matching approach may result in synchronized increases or decreases in the duration between phenological stages, thus failing to capture the true phenological dynamics. Second, the method exhibits a certain degree of latency, meaning that it is unable to capture subtle changes in vegetation growth in a timely manner. Therefore, future research could adopt intraseasonal estimation methods to address these issues and improve the model's accuracy in describing vegetation phenological dynamics. The spatiotemporal shape model fitting (STSMF) technique introduced by Cao et al. [44] combines the strengths of SMF-S and considers the spatiotemporal consistency of crop planting activities in local regions. By integrating VI time-series from nearby pixels in previous years, this approach can accurately detect intraseasonal phenology. However, STSMF still faces two main limitations in intra-seasonal detection. First, the method requires the clear identification of adjacent pixels with similar growth trajectories to the target pixel, which relies on the accurate recognition of crop type distribution. Second, satellite data are influenced by environmental factors, such as clouds and fog, which can result in incomplete VI time-series data. Therefore, applying the effective data smoothing techniques or satellite data fusion approaches is essential for improving the accuracy and reliability of intraseasonal detection.

C. Impact of Different Strategies on Model Transferability

This section establishes ten strategies to analyze the impact of two major potential factors (time window and modeling process) on the model transfer performance. The results in Tables VIII and SIII of the Supplementary Material show that transferring more data from the target domain leads to better transfer outcomes. This is because the inclusion of more target-domain data calibrates the model, a phenomenon frequently observed in model transfer studies and highlighting the importance of prior knowledge in model transfer [52]. As winter wheat's AGB typically increases or decreases slightly during the maturation stage (see Fig. 11), the AGB values in the later growth stages are larger and exert a more pronounced influence on the model validation. Therefore, using data from the entire growth period for model validation often results in higher RMSE and nRMSE values. In the analysis of data transfer across different time windows, it was found that transferring data from the single filling stage and the third stage (filling and maturation stages) produced better model

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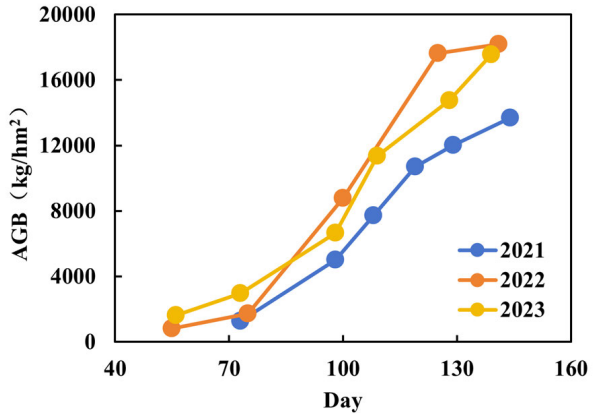


Fig. 11. Variation of winter wheat AGB with growth stages over three years of field experiments.

accuracy, suggesting that the later growth stages of winter wheat exert a more significant influence on the model.

In data transfer, the sequence of steps is often overlooked. The traditional method involves combining data from the source and target domains for model development, and then applying the resulting model to the target domain dataset. This study explores the effects of different modeling processes through three distinct strategies (see Table VIII). The results indicate that models involving "double data transfer" outperform those involving "single data transfer." This is because, during the initial data transfer, an initial calibration is performed between the source and target domains, but a cubic polynomial function cannot fit all the data points perfectly. Therefore, further double transfer and recalibration of the model output with the target-domain data significantly enhances the fitting accuracy between the source and target domains. This demonstrates that double data transfer can optimize the model's alignment with the target domain more effectively, thereby improving the model's transferability.

D. Transfer From UAV to Satellite

Although remote sensing technology is a key tool for large-scale estimation and enables continuous observation of AGB changes across vast landscapes using satellite imagery, it still encounters two limitations when estimating the AGB of winter wheat at the satellite scale. First, it is difficult to match the size of ground sampling plots with satellite pixel sizes [74]. The conventional approach involves collecting AGB data within sampling plots during field surveys, using the average value from multiple plots as a representative measure [18]. However, this disparity in spatial scales introduces significant uncertainty. Second, sample size substantially impacts the accuracy of modeling, and it is extremely challenging to obtain a large number of ground-measured data points that match satellite timing across a region [74].

Several studies have explored the use of UAV data as a bridge to resolve the disparity between ground-based and satellite data. For instance, Chen et al. [18] combined UAV and satellite data to map AGB and shrub dominance across scales in temperate grasslands. Hu et al. [19] created high-resolution maps of canopy cover in Chinese grasslands using extensive UAV imagery and satellite data. These studies first inverted shrub and herbaceous AGB at the UAV scale or calculated grassland coverage and then used the results from UAV imagery as a training dataset for the satellite scale. This method aims to close the scale gap between ground-based sampling and satellite image pixels while alleviating the workload of basic tasks. However, this approach requires a large volume of UAV imagery data, demanding considerable human and material resources, and it could introduce significant errors from UAV-scale estimations that may affect satellite-scale models. Despite these challenges, this approach has made significant progress compared with traditional satellite estimation methods.

In contrast, the method proposed in this study does not require a large number of UAV images to be obtained that correspond to the satellite data. Instead, it efficiently transfers models from UAV to satellite through the phenological matching and data transfer methods. This innovative approach effectively addresses the issue of scale matching and reduces the demand for resources. Dynamic tracking of winter wheat AGB fluctuations at the regional level is crucial for yield forecasting, agricultural management, policy formulation, and evaluating economic benefits.

E. Limitations and Future Prospects

Phenology-driven models are capable of continuously estimating the dynamic changes in AGB of winter wheat. However, their performance is significantly influenced by the functions: $\mathrm{Fun}(X)_k$ and $\mathrm{Fun}(X)_b$, as well as VI. The functions $\mathrm{Fun}(X)_k$ and $\mathrm{Fun}(X)_b$ are primarily determined by the dataset used, and different datasets can lead to substantial variations in these functions. In many cases, $\mathrm{Fun}(X)_k$ and $\mathrm{Fun}(X)_b$ cannot fully capture the relationships between all VIs and AGB within a dataset. Therefore, how to make $\mathrm{Fun}(X)_k$ and $\mathrm{Fun}(X)_b$ more reasonably represent the relationship between VI and AGB is an urgent issue to be addressed.

In phenology-driven models, the selection of VI is also a critical factor. The results in Tables IX and XI indicate that SAVI achieves relatively higher accuracy, likely because during the early growth stages of winter wheat (e.g., green-up and tillering stages), canopy coverage is low, exposing substantial soil background to the remote sensor's field of view. Soil reflectance can significantly interfere with the accuracy of VIs. SAVI mitigates this issue by introducing a soil adjustment factor L to offset and correct the Red and NIR reflectance values, thereby effectively reducing the influence of soil brightness variations on the index. Moreover, as AGB increases, leaf area and canopy coverage of winter wheat also increase, leading to a decrease in red reflectance and an increase in NIR reflectance. Consequently, VIs based on the difference $R_{\mathrm{nir}} - R_{\mathrm{red}}$ can effectively capture AGB dynamics. However, in the formulation of WDRVI, the difference term is $0.1 \times R_{\mathrm{nir}} - R_{\mathrm{red}}$, which narrows the dynamic range of the difference and thereby reduces its sensitivity and estimation capability for AGB. The results also indicate that high-phenological matching accuracy does not necessarily lead to high-AGB estimating accuracy. Consequently, future studies should delve

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deeper into how VI selection affects model accuracy and identify a more effective strategy for selecting VIs to improve the robustness and generalizability of phenology-driven models.

For the SMF and the new phenological detection method, since their primary function is phenological stage detection, the focus is mainly on the parts of the VI time-series curve that are related to phenological changes (i.e., changes associated with the x-axis). However, the VI time-series curves shown in Fig. 10 indicate that there are not only differences along the x-axis (time axis) but also significant differences along the y-axis (VI values) between the UAV and satellite scales. According to the research by Yu et al. [75], the AGB of winter wheat can be expressed by the following equation:

$$\mathrm{AGB} = \sum_{t=1}^{n} \mathrm{GPP}_t \times \mathrm{CUE} \times F \tag{12}$$

where $\sum_{t=1}^{n} \mathrm{GPP}_t$ represents the AGPP of winter wheat throughout the entire growing season (from day 1 to day n). Carbon use efficiency (CUE) is applied to convert GPP into net primary productivity (NPP) and is generally set to 0.533 [76]. F represents the conversion factor from carbon to biomass, which is 2.22 for winter wheat.s In addition, research by Badgley et al. [56] has shown that the NIRv can be used to estimate GPP. Therefore, can we rewrite (12) to (13)

$$\mathrm{AGB} = \sum_{t=1}^{n} \mathrm{NIRv}_t \times \mathrm{CUE} \times F. \tag{13}$$

For AGB estimation, AGB is an accumulative metric. Whether the differences in the VI time-series curve along the y-axis would affect the accumulation of GPP and thereby significantly impact the AGB estimation results remains unclear. Therefore, future research should not only focus on the changes of the VI time-series curve along the x-axis (time axis) but also delve into the impact of differences along the y-axis (VI values) on the model. Specifically, it can be explored whether the differences in the AGB estimation range of the model can be directly calibrated through the changes in the VI time-series curve, thereby replacing some of the work of data transfer and further improving the applicability and accuracy of the model.

VI. Conclusion

This study successfully transferred the estimation of AGB from the UAV scale to the satellite scale by integrating phenology-driven models, phenological matching, and data transfer methods. At the UAV scale, phenological matching significantly enhanced the performance of models using most VIs. Specifically, for the SAVI, the R^2 increased by 4.76%, and the nRMSE decreased by 12.12% after phenological matching. Furthermore, applying a two-step data transfer approach and focusing on data from the later growth stages led to further improvements in model accuracy. When the optimized UAV model was used at the satellite scale, SAVI reached an R^2 of 0.68 and an RMSE of 2552.17 kg/hm^2 following phenological matching. After conducting data transfer in the final growth stage, the model's R^2 increased to 0.75, and RMSE reduced to 2279.71 kg/hm^2. These results indicate that the methods proposed in this study can effectively transfer UAV models to the satellite scale, demonstrating the significant potential for application.

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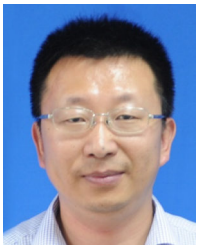
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